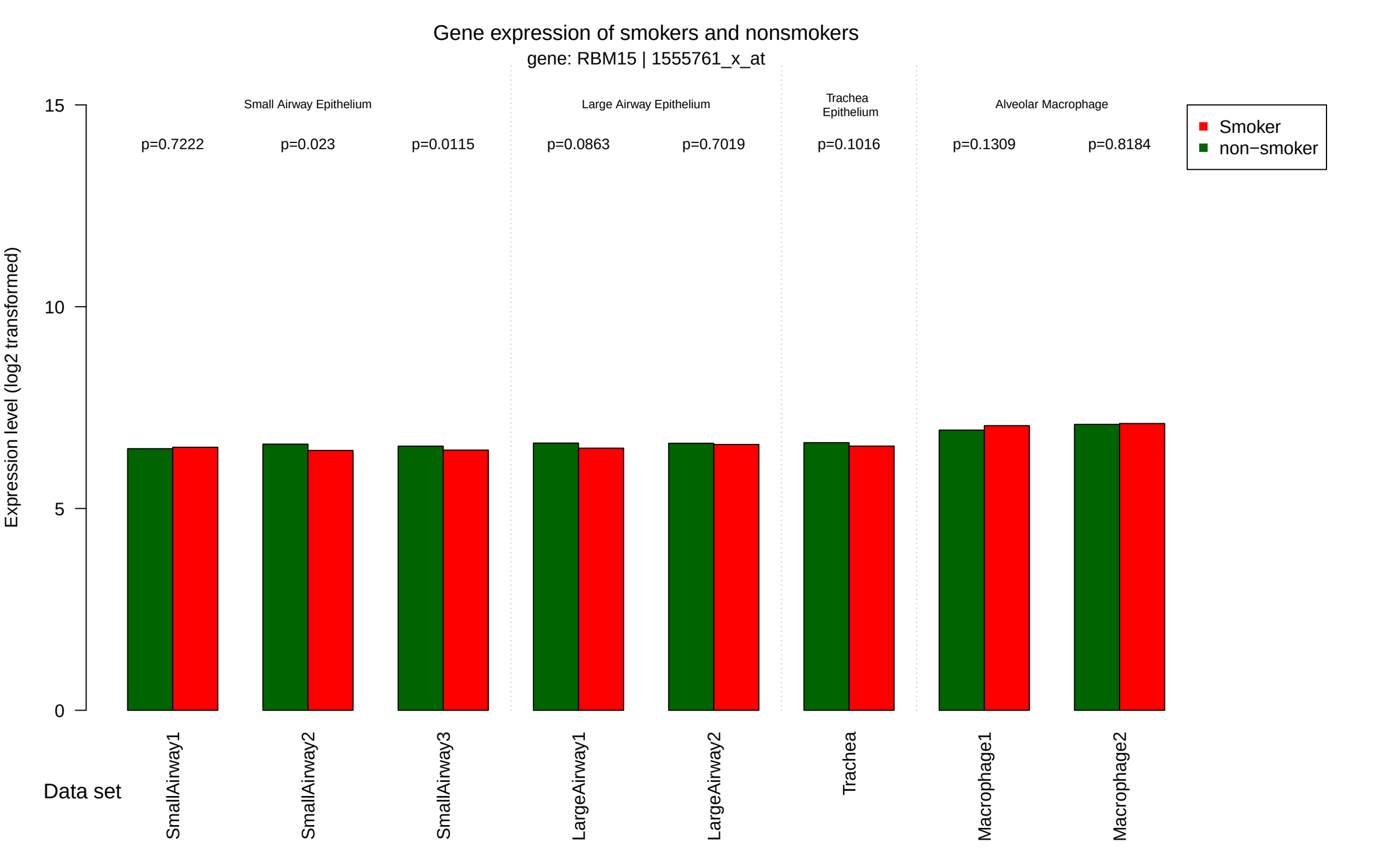
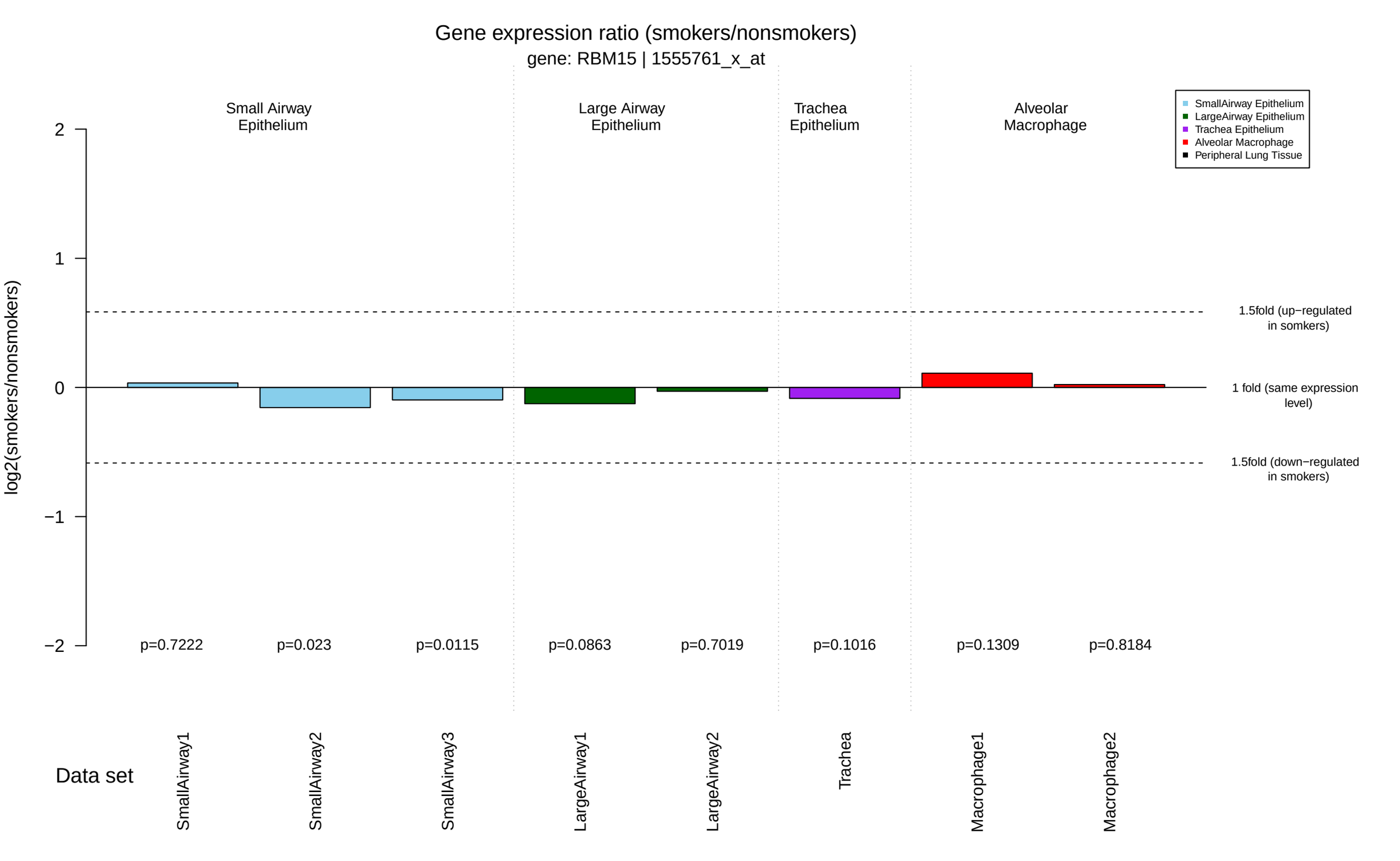
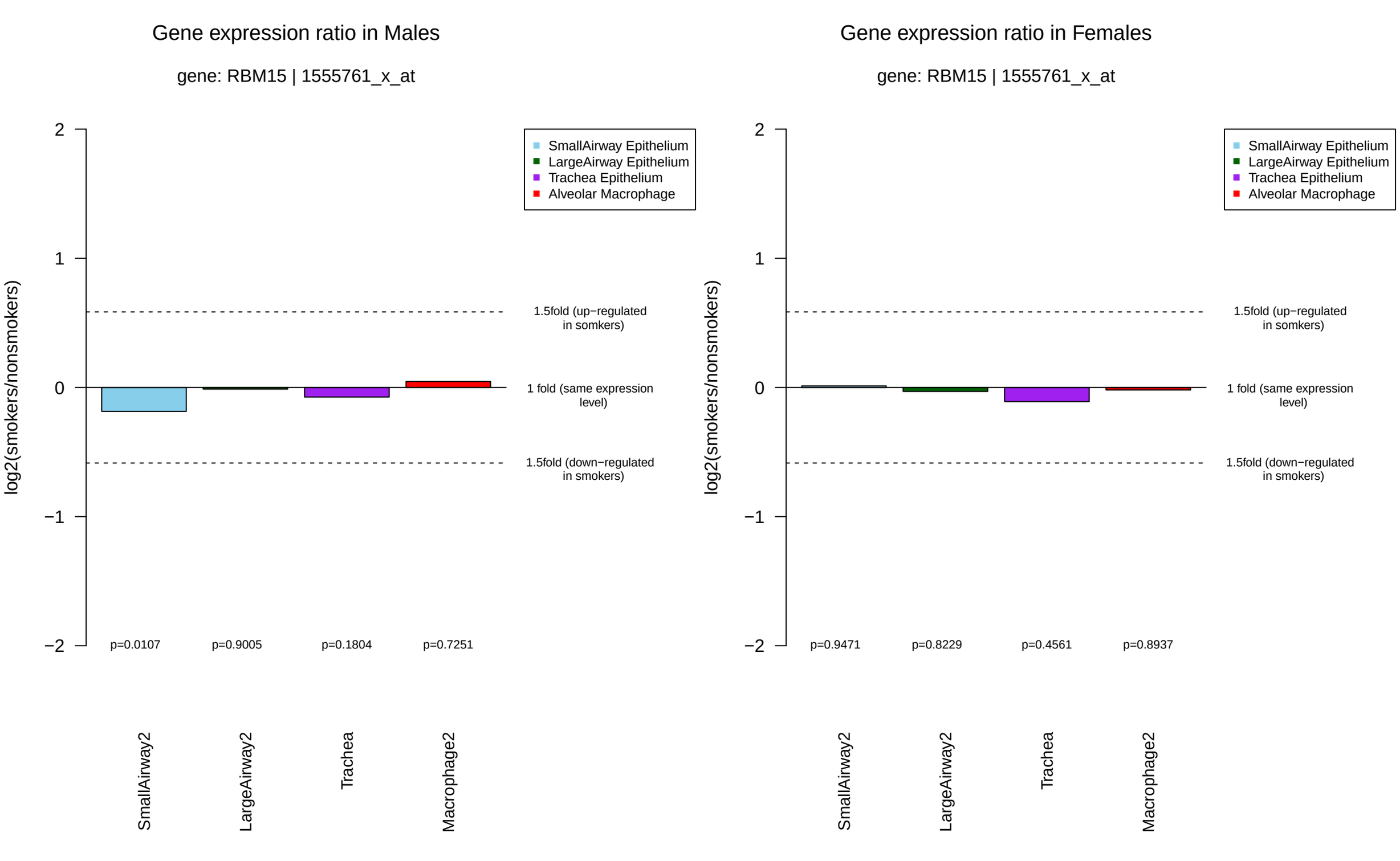
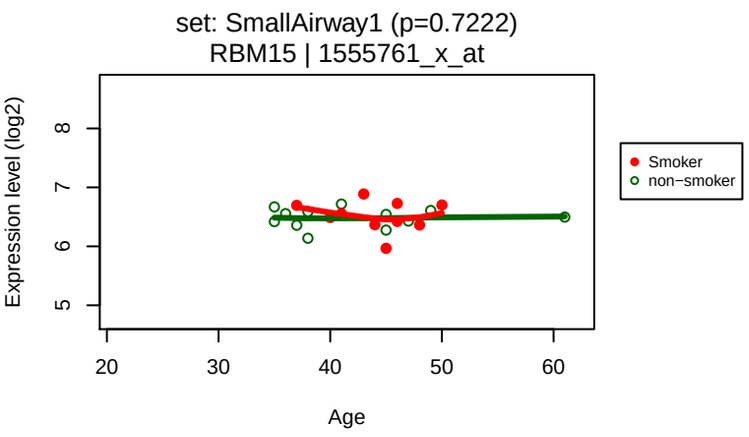
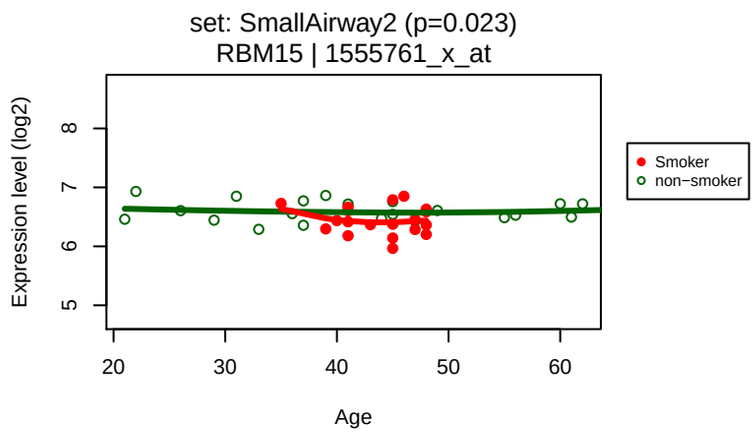
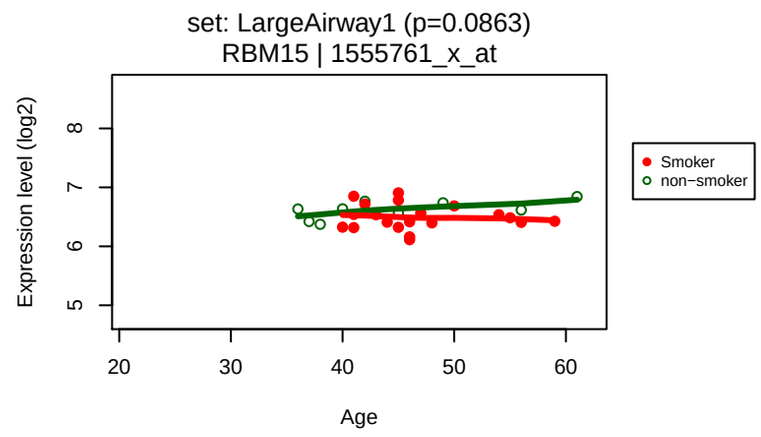

Gene expression of smokers and nonsmokers
gene: RBM15 | 1555761_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7222
p=0.023
p=0.0115
p=0.0863
p=0.7019
p=0.1016
p=0.1309
p=0.8184
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

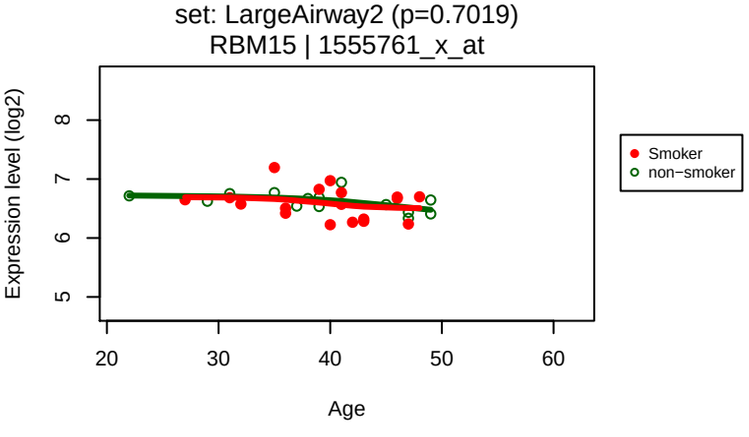

Gene expression ratio (smokers/nonsmokers)
gene: RBM15 | 1555761_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7222
p=0.023
p=0.0115
p=0.0863
p=0.7019
p=0.1016
p=0.1309
p=0.8184
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

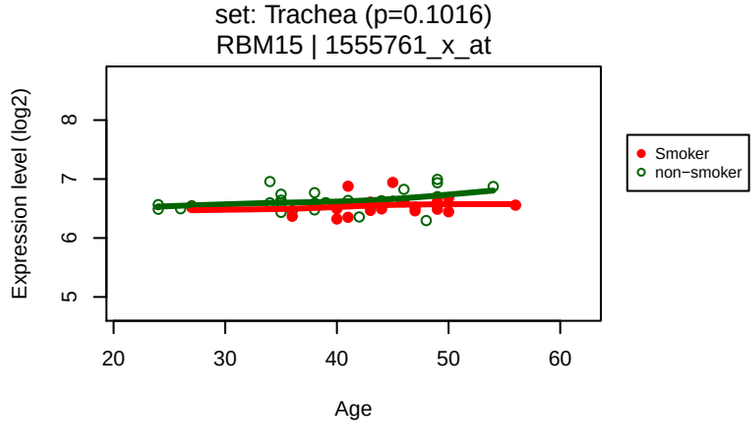

Gene expression ratio in Males
gene: RBM15 | 1555761_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0107
p=0.9005
p=0.1804
p=0.7251
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM15 | 1555761_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9471
p=0.8229
p=0.4561
p=0.8937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

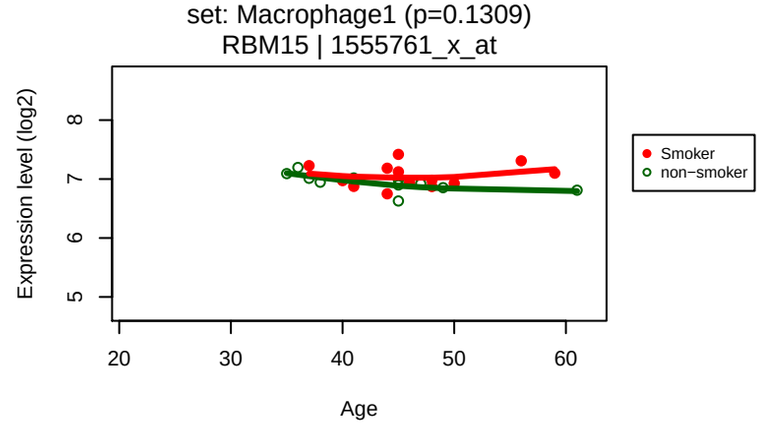

set: SmallAirway1 (p=0.7222)
RBM15 | 1555761_x_at
Expression level (log2)
Age
Smoker
non−smoker

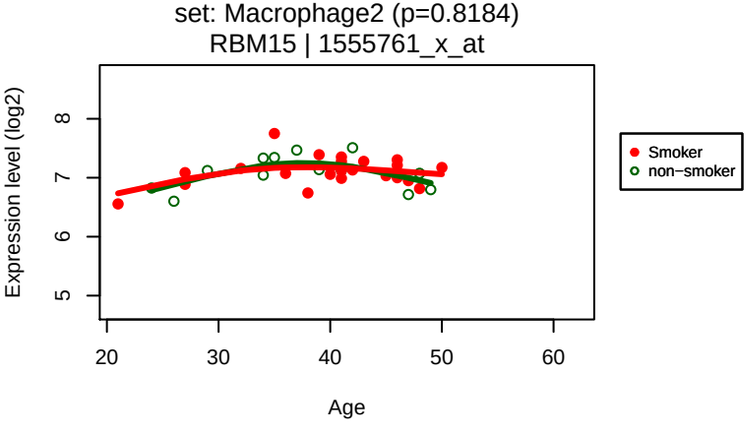

set: SmallAirway2 (p=0.023)
RBM15 | 1555761_x_at
Expression level (log2)
Age
Smoker
non−smoker

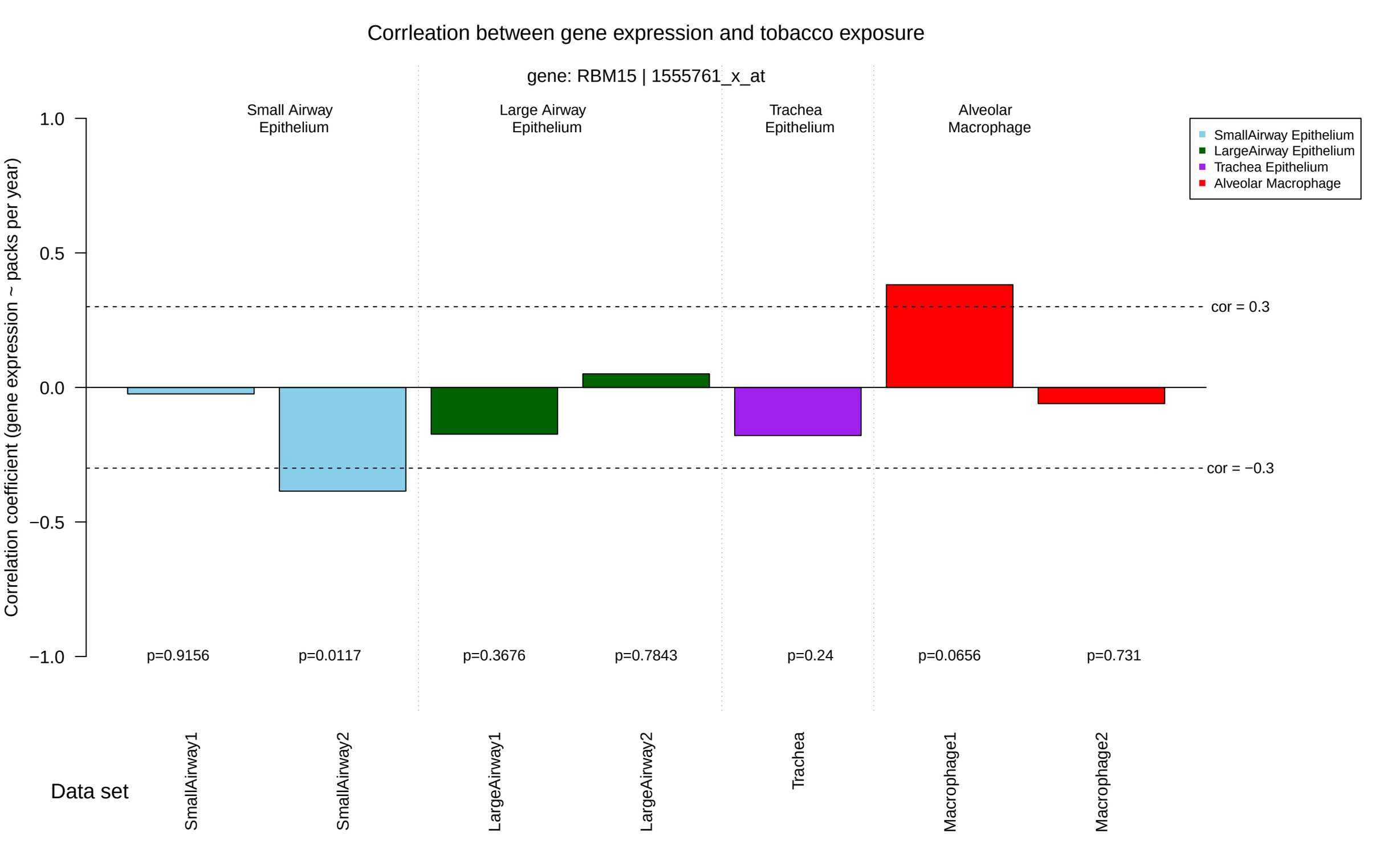

set: LargeAirway1 (p=0.0863)
RBM15 | 1555761_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7019)
RBM15 | 1555761_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1016)
RBM15 | 1555761_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1309)
RBM15 | 1555761_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8184)
RBM15 | 1555761_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM15 | 1555761_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9156
p=0.0117
p=0.3676
p=0.7843
p=0.24
p=0.0656
p=0.731
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2